Supplementary Material C

Application Supplement to "Latent Spatial Regimes in Spatio-Temporal State-Space Models, with an Application to Local Price Leadership in Fuel Markets"

This supplement documents the application of Section 5 of the manuscript: local price leadership among the fuel stations of eleven Italian metropolitan cities. Section C1 describes the data, their pre-treatment and the three settings; Section C2 the model, its estimation and its cost; Section C3 every selected model; Section C4 every regime, with the tests of leadership and the long-run pass-through; Section C5 the pump-by-pump benchmark; Section C6 the diagnostics and the checks of robustness; Section C7 the pass-through of the excise duties; Section C8 the small samples of the independent stations; Section C9 the replication material.

C1 The data and the three settings

The source. The prices are those of Osservaprezzi Carburanti, the platform of the Italian ministry on which every station posts its prices (Demir, 2026). The file is a long panel, one row per station and day, from 1 January 2016 to 30 June 2026, with the self-service and full-service prices of gasoline and diesel, the brand, the type of station, the municipality, the province and the coordinates; prices appear to be recorded at each change and carried forward. The application uses the self-service prices.

The window. From Monday 3 January 2022 to Sunday 28 June 2026: 1,638 days, 234 weeks. The window starts after the two holes of the data (July to September 2018, the second half of June 2021), contains every fiscal event of Section C7, and starts with the change in pricing behavior of 2022, when the share of station-days without a price change fell from about 92% to between 75% and 87%. The 52 weeks of 2021 are a presample, used for the lags only.

The eleven cities. A metropolitan city enters if it keeps at least ten independent stations, so that the settings that involve them can be estimated: Naples (71 independents), Rome (50), Catania (25), Turin and Bologna (23), Milan (17), Palermo and Messina (14) and Bari (13). Florence (6) and Venice (8) are added by choice, for their weight among Italian cities; Genoa and Cagliari keep no independent station. The stations kept, by city:

	Rome	Milan	Turin	Naples	Bari	Bologna	Catania	Florence	Palermo	Venice	Messina
stations	653	423	419	339	240	208	202	189	182	163	118
independent	50	17	23	71	13	23	25	6	14	8	14

3,136 stations in all. Prices are sticky even at weekly frequency: the Sunday price is unchanged from the previous week in 25 to 36% of the weeks for gasoline and 21 to 31% for diesel (medians by city), most in Palermo and Naples.

Relative prices. Let p_{it} be the Sunday price of a fuel at station i in week t, in cents per liter. The relative price is

$$z_{it} = (p_{it} - \bar{p}_t) - \frac{1}{T}\sum_{s=1}^{T}(p_{is} - \bar{p}_s), \tag{C1}$$

with $\bar{p}_t$ the mean over all the non-highway stations of Italy that report the fuel in week t, some 20,000. The first step removes the stochastic trend of costs, of the exchange rate and of the part of every change of the excise duties that all stations pass through at once; it uses the values of week t only, so the temporal order that a Granger test reads is intact. The second removes the permanent premium of the station, the brand premium above all. The national mean is preferred to the mean of the city because the deviations from the city mean sum to zero within the city, which would make the latent processes of the regimes linearly dependent; the weight of a city in the national mean is below 1%. The relative prices revert to their mean: the augmented Dickey–Fuller test rejects a unit root for 96 to 100% of the stations of every city, both fuels, and their median first-order autocorrelation is 0.52 to 0.71.

The settings. The neighbor X of a station Y is its nearest station of a given group: in setting 1, Y is a station of a major brand (Eni, Api-Ip, Esso, Q8, Tamoil) and X the nearest independent station; in setting 2, Y is independent and X the nearest station of a major brand; in setting 3, Y is a station of a major brand and X the nearest station of another major brand. Stations of the other brands are in no group. Settings 1 and 3 have 2,427 stations Y, setting 2 has 264. Distances are straight lines: the nearest independent of a major-brand station lies at a median of 1.4 kilometers in Naples and 2.8 to 7.9 kilometers elsewhere, and only 12% of the major-brand stations have an independent within about one kilometer; the nearest station of another major brand lies at a median of 0.6 to 1.6 kilometers. Setting 1 is therefore mostly a relation between stations of the same city rather than between immediate neighbors, and the distance of X is reported with every regime (Section C4). With 11 cities and 2 fuels, the three settings make 66 models.

C2 The model and its estimation

The model. For a station i in regime k, with z_{it} its relative price of the fuel, o_{it} its relative price of the other fuel, and x_{it}, w_{it} those of the two fuels at its neighbor,

$$z_{it} = \beta_{0k} + \sum_{l=1}^{4}\left(b_{kl}z_{i,t-l} + c_{kl}o_{i,t-l} + d_{kl}x_{i,t-l} + e_{kl}w_{i,t-l}\right) + \gamma_k'\pi_t + y_t^{(k)} + \epsilon_{it}, \tag{C2}$$

the model of Section 5.3 of the manuscript: π_t collects ten pulses, the week of each of the five changes of the excise duties and the week after; $y_t^{(k)}$ is the latent AR(1) process of the regime; ϵ_{it} the measurement error with a nugget and an exponential spatial correlation. A model has 27 regression coefficients and 33 parameters per regime.

The lags. The four blocks take lags 1 to 4. The partial autocorrelation function of the relative price, pump by pump, exceeds the band $\pm 2/\sqrt{T}$ for all the pumps at lag 1, for 56 to 72% at lag 2, 26 to 41% at lag 3 and 14 to 22% at lag 4 (medians over the cities), and for at most 17% from lag 5 on; the order of the autoregression chosen by the AIC pump by pump has median 3 to 4 and quartiles 2 and 5 to 6. The partial autocorrelation at lag 52 is within the band for 99 to 100% of the pumps. One choice of lags serves every setting, city and fuel. The missing weeks of the lagged

series (1.4 to 2.1% of the station-weeks) are completed by the Kalman smoother of a pooled STEM model of the city with the intercept and the pulses; the response keeps its missing weeks.

Selection and inference. The grid has $K = 1, \dots, K_{\max}$ with $K_{\max} = \min(5, \lfloor n/5 \rfloor)$ for n stations, the pooled model below ten, and $\phi \in \{0, 0.025, 0.05, 0.1, 0.2, 0.5, 1\}$, up to 29 fits per model; $(\hat{K}, \hat{\phi})$ follow from the two-step rule on the BIC with the band $[0.025, 0.2]$, the minimum regime size is five and the Potts graph has five neighbors. The refit-with-clustering bootstrap of the selected fit and of the pooled model has $B = 100$ draws. The Wald test of a block of q coefficients in regime k is $W_k = \hat{b}_k' \hat{V}_k^{-1} \hat{b}_k$, with $\hat{V}_k$ the covariance of the aligned bootstrap draws, against the chi-square on q degrees of freedom; the long-run pass-through $\lambda_k = \sum_l d_{kl} / (1 - \sum_l b_{kl})$ and its percentile interval come from the same draws.

C3 The selected models

Table C1 lists the 66 selected models. In settings 1 and 3 every model has four or five regimes, five, the cap, in 38 of 44; in setting 2 the cap $\lfloor n/5 \rfloor$ binds in 13 of 22. The penalty is small: $\hat{\phi} = 0$ in 29 models, within the band in 35, and 0.5 and 1 in two models of setting 2 with fewer than twenty stations. The adjusted Rand index between the regimes and the brands of the stations is 0.07 to 0.48 (median 0.23); in 25 of the 44 models of settings 1 and 3 one regime is made of Eni stations only, 19 to 96 of them, and in two of Q8 stations only. Every bootstrap has 100 usable draws out of 100, and the partition re-estimated on them coincides with the estimated one in the median draw of every model but four: the independents of Naples for gasoline (median adjusted Rand index 0.66), and Catania diesel (0.90), Rome diesel (0.96) and Rome gasoline (0.99) in setting 3.

Table C1: The selected models. $K_{\max}$, the largest number of regimes of the grid; $\hat{K}$, $\hat{\phi}$ and the sizes of the regimes of the selected fit; ARI brands, the adjusted Rand index between the regimes and the brands of the stations Y (not defined in setting 2, where they are all independent); usable draws, of $B = 100$; ARI draws, the median adjusted Rand index between the partitions of the bootstrap draws and the estimated one.

setting	city	fuel	pumps	$K_{\max}$	$\hat{K}$, $\hat{\phi}$, sizes	ARI brands	usable draws	ARI draws
1	Turin	diesel	332	5	5, 0.025, 40/69/58/92/73	0.47	100	1.00
1	Turin	gasoline	332	5	5, 0.050, 39/149/60/56/28	0.15	100	1.00
1	Milan	diesel	368	5	5, 0.000, 23/125/72/55/93	0.29	100	1.00
1	Milan	gasoline	368	5	4, 0.025, 54/104/33/177	0.17	100	1.00
1	Venice	diesel	100	5	5, 0.100, 24/18/7/25/26	0.45	100	1.00
1	Venice	gasoline	100	5	5, 0.025, 9/17/20/18/36	0.12	100	1.00
1	Bologna	diesel	156	5	4, 0.050, 29/10/50/67	0.25	100	1.00
1	Bologna	gasoline	156	5	5, 0.000, 14/39/17/44/42	0.14	100	1.00
1	Florence	diesel	143	5	5, 0.025, 30/40/6/57/10	0.25	100	1.00
1	Florence	gasoline	143	5	5, 0.000, 55/18/29/13/28	0.26	100	1.00
1	Rome	diesel	551	5	4, 0.025, 61/113/96/281	0.23	100	1.00
1	Rome	gasoline	551	5	5, 0.025, 127/107/66/176/75	0.18	100	1.00
1	Naples	diesel	231	5	5, 0.000, 34/55/40/57/45	0.33	100	1.00
1	Naples	gasoline	231	5	5, 0.000, 29/51/63/63/25	0.16	100	1.00
1	Bari	diesel	200	5	4, 0.100, 79/17/44/60	0.48	100	1.00
1	Bari	gasoline	200	5	5, 0.200, 55/79/12/5/49	0.45	100	1.00
1	Palermo	diesel	147	5	5, 0.025, 26/54/39/18/10	0.12	100	1.00
1	Palermo	gasoline	147	5	5, 0.050, 41/27/9/44/26	0.19	100	1.00
1	Catania	diesel	108	5	5, 0.000, 8/40/28/26/6	0.23	100	1.00
1	Catania	gasoline	108	5	4, 0.000, 29/8/37/34	0.14	100	1.00

setting	city	fuel	pumps	$K_{\max}$	$\hat{K}$, $\hat{\phi}$, sizes	ARI brands	usable draws	ARI draws
1	Messina	diesel	91	5	5, 0.000, 32/20/5/20/14	0.30	100	1.00
1	Messina	gasoline	91	5	5, 0.000, 32/5/7/22/25	0.22	100	1.00
2	Turin	diesel	23	4	3, 0.000, 10/5/8	–	100	1.00
2	Turin	gasoline	23	4	4, 0.000, 7/5/6/5	–	100	1.00
2	Milan	diesel	17	3	3, 0.500, 6/5/6	–	100	1.00
2	Milan	gasoline	17	3	2, 0.000, 12/5	–	100	1.00
2	Venice	diesel	8	1	1, 0.000, 8	–	100	1.00
2	Venice	gasoline	8	1	1, 0.000, 8	–	100	1.00
2	Bologna	diesel	23	4	3, 0.000, 13/5/5	–	100	1.00
2	Bologna	gasoline	23	4	3, 0.000, 5/13/5	–	100	1.00
2	Florence	diesel	6	1	1, 0.000, 6	–	100	1.00
2	Florence	gasoline	6	1	1, 0.000, 6	–	100	1.00
2	Rome	diesel	50	5	5, 0.025, 20/9/5/5/11	–	100	1.00
2	Rome	gasoline	50	5	5, 0.200, 16/15/8/5/6	–	100	1.00
2	Naples	diesel	71	5	4, 0.000, 37/12/17/5	–	100	1.00
2	Naples	gasoline	71	5	4, 0.100, 28/5/10/28	–	100	0.66
2	Bari	diesel	13	2	2, 0.000, 5/8	–	100	1.00
2	Bari	gasoline	13	2	2, 1.000, 5/8	–	100	1.00
2	Palermo	diesel	14	2	1, 0.000, 14	–	100	1.00
2	Palermo	gasoline	14	2	2, 0.000, 9/5	–	100	1.00
2	Catania	diesel	25	5	4, 0.000, 6/6/5/8	–	100	1.00
2	Catania	gasoline	25	5	4, 0.000, 5/9/6/5	–	100	1.00
2	Messina	diesel	14	2	2, 0.000, 9/5	–	100	1.00
2	Messina	gasoline	14	2	2, 0.025, 6/8	–	100	1.00
3	Turin	diesel	332	5	5, 0.100, 37/56/135/45/59	0.29	100	1.00
3	Turin	gasoline	332	5	5, 0.050, 49/52/123/41/67	0.17	100	1.00
3	Milan	diesel	368	5	5, 0.000, 103/33/74/97/61	0.39	100	1.00
3	Milan	gasoline	368	5	5, 0.000, 74/35/103/102/54	0.29	100	1.00
3	Venice	diesel	100	5	5, 0.025, 32/15/22/23/8	0.30	100	1.00
3	Venice	gasoline	100	5	5, 0.200, 37/12/10/21/20	0.23	100	1.00
3	Bologna	diesel	156	5	5, 0.000, 28/6/51/32/39	0.28	100	1.00
3	Bologna	gasoline	156	5	5, 0.100, 41/22/19/64/10	0.17	100	1.00
3	Florence	diesel	143	5	5, 0.200, 8/38/56/8/33	0.31	100	1.00
3	Florence	gasoline	143	5	5, 0.025, 12/52/16/35/28	0.22	100	1.00
3	Rome	diesel	551	5	5, 0.025, 62/177/85/200/27	0.32	100	0.96
3	Rome	gasoline	551	5	5, 0.025, 55/215/135/5/141	0.09	100	0.99
3	Naples	diesel	231	5	5, 0.100, 32/58/51/39/51	0.24	100	1.00
3	Naples	gasoline	231	5	5, 0.025, 29/24/62/50/66	0.14	100	1.00
3	Bari	diesel	200	5	5, 0.100, 51/26/44/5/74	0.45	100	1.00
3	Bari	gasoline	200	5	5, 0.200, 75/5/53/13/54	0.45	100	1.00
3	Palermo	diesel	147	5	4, 0.050, 67/45/13/22	0.07	100	1.00
3	Palermo	gasoline	147	5	5, 0.200, 29/8/37/41/32	0.20	100	1.00
3	Catania	diesel	108	5	5, 0.100, 26/10/5/28/39	0.23	100	0.90
3	Catania	gasoline	108	5	5, 0.000, 40/23/5/8/32	0.15	100	1.00
3	Messina	diesel	91	5	5, 0.200, 9/14/20/23/25	0.37	100	1.00
3	Messina	gasoline	91	5	5, 0.050, 16/26/16/28/5	0.27	100	1.00

C4 Leadership by regime

Tables C2 to C4 report every regime: its size, its most frequent brand with its share, the median distance from its stations Y to their neighbors X, the p-values of the bootstrap Wald tests of the four hypotheses — the eight lags of X (the leadership test of the manuscript), the four lags of X on the same fuel, the four on the other fuel, and the four lags of the other fuel of Y — and the long-run pass-through $\hat{\lambda}_k$ with its bootstrap 95% interval.

The lags of X, fuel by fuel. The neighbor's price of the same fuel carries most of the leadership: its four lags are significant at 5% in 47, 37 and 67% of the regimes of settings 1 to 3, those of the other fuel in 33, 25 and 43%. The other fuel of the same station predicts its price in 81% of the regimes.

Brands. The regimes in which Eni stations are the majority, 24 in setting 1 and 25 in setting 3, follow their neighbor less often and less:

setting	regimes	number	X leads (%)	median $\hat{\lambda}_k$
1	Eni $>$ 50%	24	50	0.04
1	other	81	73	0.07
3	Eni $>$ 50%	25	72	0.14
3	other	84	86	0.17

All the regimes at once. The hypothesis that X leads in no regime of a model is tested by stacking the coefficients of the K regimes, with their joint bootstrap covariance, on $8K$ degrees of freedom (Table C5). It is rejected in 65 of the 66 models, against the chi-square and against the bootstrap distribution of the statistic centered at the estimate; the pooled model rejects in 51.

The regimes in space. Figure 4 of the manuscript shows two partitions of setting 3; Figure C1 adds Rome for setting 1 and Naples for setting 3. In every city the regimes interpenetrate.

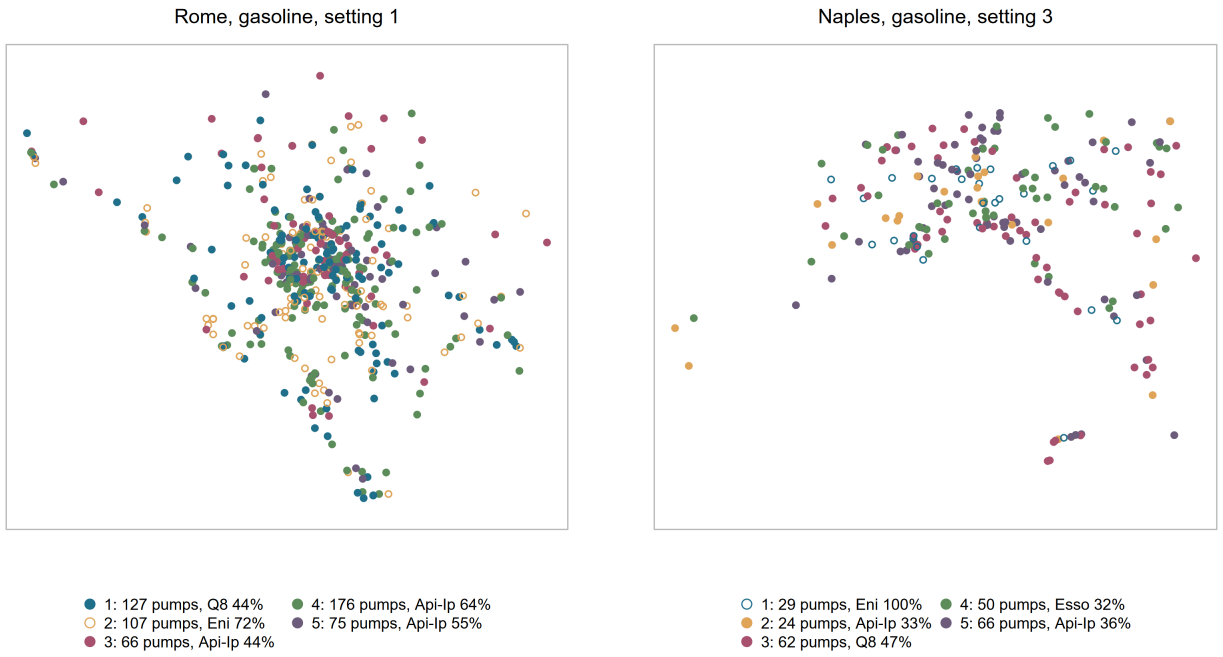


Figure C1: The regimes of Rome, gasoline, setting 1 (left) and of Naples, gasoline, setting 3 (right). Every station Y is colored by its regime; filled marks: the regime where the neighbor leads, open marks: where it does not; the legend gives the size of each regime and its most frequent brand with its share.

Setting 1: Y a station of a major brand, X the nearest independent

Table C2: The regimes of setting 1. Brand: the most frequent brand of the stations Y of the regime, with its share; X: median distance to the neighbor, in kilometers; p-values of the bootstrap Wald test of the eight lags of X, of its four lags on the same fuel, of its four lags on the other fuel, and of the four lags of the other fuel of Y; long-run pass-through from X on the same fuel, with its bootstrap 95% interval.

city	fuel	regime	pumps	brand (%)	X (km)	bootstrap Wald, p-value				long-run pass-through
						X	X same	X other	Y other	from X, same fuel
Turin	diesel	1	40	Tamoil 95	6.2	0.018	0.006	0.316	<0.001	0.09 (0.02, 0.16)
Turin	diesel	2	69	Api-Ip 32	4.7	<0.001	<0.001	0.072	<0.001	0.24 (0.13, 0.35)
Turin	diesel	3	58	Eni 100	4.0	0.004	0.341	0.002	<0.001	0.01 (−0.10, 0.10)
Turin	diesel	4	92	Q8 71	3.5	0.360	0.527	0.387	<0.001	0.06 (−0.01, 0.11)
Turin	diesel	5	73	Api-Ip 73	5.0	<0.001	<0.001	0.055	<0.001	0.12 (0.05, 0.17)
Turin	gasoline	1	39	Eni 100	4.1	0.094	0.042	0.192	<0.001	0.17 (0.06, 0.29)
Turin	gasoline	2	149	Api-Ip 34	3.9	0.001	0.001	0.297	<0.001	0.09 (0.03, 0.15)
Turin	gasoline	3	60	Tamoil 55	5.7	<0.001	0.004	0.205	<0.001	0.12 (0.02, 0.24)
Turin	gasoline	4	56	Q8 41	4.0	0.074	0.050	0.247	<0.001	0.02 (−0.05, 0.10)
Turin	gasoline	5	28	Api-Ip 39	6.6	<0.001	0.021	<0.001	0.006	0.10 (−0.23, 0.40)
Milan	diesel	1	23	Api-Ip 35	4.5	0.003	0.016	0.214	<0.001	−0.40 (−0.73, −0.05)
Milan	diesel	2	125	Q8 44	3.9	<0.001	0.003	0.231	<0.001	0.00 (−0.06, 0.06)
Milan	diesel	3	72	Eni 100	4.1	0.031	0.123	0.004	<0.001	−0.16 (−0.33, −0.01)
Milan	diesel	4	55	Api-Ip 31	3.9	<0.001	0.501	0.306	0.539	0.04 (−0.09, 0.13)
Milan	diesel	5	93	Q8 49	3.7	<0.001	<0.001	0.130	<0.001	0.03 (−0.04, 0.10)
Milan	gasoline	1	54	Eni 100	4.0	0.064	0.082	0.161	<0.001	0.11 (−0.07, 0.25)
Milan	gasoline	2	104	Q8 41	3.7	<0.001	0.018	0.152	<0.001	0.07 (0.01, 0.14)
Milan	gasoline	3	33	Q8 30	4.4	<0.001	<0.001	0.021	<0.001	0.35 (0.03, 0.64)
Milan	gasoline	4	177	Api-Ip 37	4.0	<0.001	<0.001	0.005	<0.001	0.10 (0.04, 0.17)
Venice	diesel	1	24	Api-Ip 67	5.7	0.021	0.036	0.681	<0.001	−0.04 (−0.18, 0.09)
Venice	diesel	2	18	Esso 44	11.1	0.399	0.416	0.727	0.183	−0.02 (−0.21, 0.14)
Venice	diesel	3	7	Api-Ip 57	7.6	<0.001	0.123	0.038	0.081	−0.42 (−1.04, 0.05)
Venice	diesel	4	25	Q8 68	5.5	<0.001	<0.001	0.030	<0.001	0.03 (−0.08, 0.16)
Venice	diesel	5	26	Eni 100	11.3	<0.001	0.582	0.002	<0.001	0.09 (−0.13, 0.31)
Venice	gasoline	1	9	Api-Ip 44	7.6	<0.001	0.086	<0.001	<0.001	−0.39 (−0.70, −0.19)
Venice	gasoline	2	17	Q8 41	13.4	0.588	0.769	0.240	0.019	0.07 (−0.13, 0.26)
Venice	gasoline	3	20	Api-Ip 55	8.3	<0.001	0.016	<0.001	0.019	0.14 (−0.03, 0.27)
Venice	gasoline	4	18	Eni 89	10.3	0.029	0.016	0.698	0.299	0.11 (−0.13, 0.39)
Venice	gasoline	5	36	Q8 39	5.5	0.011	0.059	0.005	<0.001	0.07 (−0.03, 0.20)
Bologna	diesel	1	29	Eni 100	2.5	<0.001	0.643	<0.001	<0.001	−0.07 (−0.20, 0.10)
Bologna	diesel	2	10	Eni 40	3.4	0.001	0.010	0.450	<0.001	0.05 (−0.29, 0.28)
Bologna	diesel	3	50	Api-Ip 32	3.2	<0.001	<0.001	<0.001	<0.001	0.20 (0.13, 0.27)
Bologna	diesel	4	67	Q8 48	3.0	<0.001	<0.001	0.025	<0.001	0.09 (0.02, 0.16)
Bologna	gasoline	1	14	Eni 36	3.2	0.005	0.001	0.283	<0.001	−0.01 (−0.33, 0.20)
Bologna	gasoline	2	39	Eni 54	2.9	0.061	0.023	0.045	0.001	0.19 (0.07, 0.28)
Bologna	gasoline	3	17	Q8 53	7.3	0.940	0.715	0.999	0.202	0.14 (−0.18, 0.27)
Bologna	gasoline	4	44	Api-Ip 55	1.9	0.008	0.146	0.006	<0.001	0.04 (−0.06, 0.19)
Bologna	gasoline	5	42	Q8 45	3.3	<0.001	<0.001	0.674	0.002	0.19 (0.09, 0.27)
Florence	diesel	1	30	Eni 100	8.2	0.028	0.086	0.005	<0.001	−0.01 (−0.17, 0.15)
Florence	diesel	2	40	Q8 50	6.9	<0.001	<0.001	0.056	<0.001	0.02 (−0.08, 0.13)
Florence	diesel	3	6	Q8 67	12.6	0.691	0.633	0.962	0.045	0.01 (−0.38, 0.43)
Florence	diesel	4	57	Q8 39	6.5	0.001	0.004	0.068	0.458	0.18 (0.08, 0.27)
Florence	diesel	5	10	Eni 40	16.7	0.914	0.712	0.543	0.009	0.08 (−0.15, 0.30)
Florence	gasoline	1	55	Q8 38	6.3	<0.001	0.003	0.097	<0.001	0.02 (−0.10, 0.13)
Florence	gasoline	2	18	Q8 100	7.5	<0.001	<0.001	0.016	0.003	−0.06 (−0.22, 0.10)
Florence	gasoline	3	29	Tamoil 41	10.7	<0.001	0.610	<0.001	0.020	−0.01 (−0.14, 0.11)
Florence	gasoline	4	13	Q8 38	11.3	0.512	0.209	0.677	<0.001	−0.08 (−0.34, 0.15)
Florence	gasoline	5	28	Eni 100	8.2	0.270	0.326	0.169	0.096	−0.17 (−0.44, 0.09)
Rome	diesel	1	61	Api-Ip 43	2.7	0.004	0.272	0.718	<0.001	−0.05 (−0.15, 0.05)
Rome	diesel	2	113	Api-Ip 50	2.6	0.002	0.015	<0.001	<0.001	0.07 (0.03, 0.12)
Rome	diesel	3	96	Eni 100	2.8	0.999	0.987	0.991	0.021	0.09 (−0.09, 0.26)
Rome	diesel	4	281	Api-Ip 44	2.8	0.001	0.013	0.785	<0.001	0.03 (0.00, 0.05)
Rome	gasoline	1	127	Q8 44	2.6	0.027	0.076	0.198	<0.001	0.03 (−0.02, 0.07)
Rome	gasoline	2	107	Eni 72	3.8	0.445	0.885	0.872	0.062	−0.01 (−0.06, 0.13)

city	fuel	regime	pumps	brand (%)	X (km)	bootstrap Wald, p-value				long-run pass-through
						X	X same	X other	Y other	from X, same fuel
Rome	gasoline	3	66	Api-Ip 44	2.9	<0.001	0.005	0.296	<0.001	0.15 (0.05, 0.25)
Rome	gasoline	4	176	Api-Ip 64	2.8	0.011	0.046	0.380	<0.001	0.02 (−0.03, 0.06)
Rome	gasoline	5	75	Api-Ip 55	2.4	0.004	0.134	0.609	0.036	0.12 (0.04, 0.27)
Naples	diesel	1	34	Eni 32	1.3	0.008	0.001	0.012	<0.001	0.15 (0.03, 0.28)
Naples	diesel	2	55	Api-Ip 67	1.4	<0.001	0.065	0.384	0.003	−0.01 (−0.08, 0.06)
Naples	diesel	3	40	Eni 98	1.3	0.758	0.744	0.600	<0.001	0.04 (−0.05, 0.13)
Naples	diesel	4	57	Q8 51	1.4	<0.001	<0.001	0.006	0.006	0.09 (0.02, 0.14)
Naples	diesel	5	45	Q8 62	1.7	0.753	0.942	0.995	0.870	0.06 (−0.05, 0.30)
Naples	gasoline	1	29	Eni 100	1.4	0.937	0.765	0.961	<0.001	0.04 (−0.09, 0.17)
Naples	gasoline	2	51	Api-Ip 45	1.1	0.282	0.282	0.698	0.107	0.16 (0.04, 0.35)
Naples	gasoline	3	63	Q8 46	1.4	<0.001	<0.001	0.222	<0.001	0.01 (−0.05, 0.07)
Naples	gasoline	4	63	Q8 43	1.6	<0.001	<0.001	0.532	0.005	0.06 (0.01, 0.14)
Naples	gasoline	5	25	Esso 32	1.2	<0.001	0.002	<0.001	<0.001	0.21 (0.06, 0.32)
Bari	diesel	1	79	Q8 54	6.8	0.320	0.071	0.251	<0.001	0.05 (−0.02, 0.13)
Bari	diesel	2	17	Api-Ip 41	7.6	0.479	0.298	0.302	<0.001	0.00 (−0.56, 0.46)
Bari	diesel	3	44	Eni 100	5.3	0.002	0.455	0.122	0.037	0.10 (0.02, 0.22)
Bari	diesel	4	60	Api-Ip 95	6.9	0.627	0.546	0.826	0.025	−0.03 (−0.08, 0.03)
Bari	gasoline	1	55	Eni 95	5.8	0.200	0.105	0.830	<0.001	0.00 (−0.06, 0.06)
Bari	gasoline	2	79	Q8 52	6.5	0.103	0.649	0.364	<0.001	−0.01 (−0.09, 0.06)
Bari	gasoline	3	12	Api-Ip 50	7.8	0.007	0.231	0.004	<0.001	−0.70 (−2.84, 0.65)
Bari	gasoline	4	5	Api-Ip 60	6.1	<0.001	0.218	0.041	<0.001	0.43 (−0.02, 0.84)
Bari	gasoline	5	49	Api-Ip 98	7.1	0.150	0.234	0.059	0.002	0.06 (0.00, 0.14)
Palermo	diesel	1	26	Eni 96	3.8	0.555	0.814	0.390	<0.001	0.04 (−0.06, 0.14)
Palermo	diesel	2	54	Eni 35	1.7	<0.001	<0.001	0.005	<0.001	0.09 (0.03, 0.16)
Palermo	diesel	3	39	Esso 36	3.4	0.511	0.331	0.832	<0.001	0.01 (−0.08, 0.09)
Palermo	diesel	4	18	Q8 89	5.5	0.017	0.028	0.113	<0.001	0.09 (−0.05, 0.23)
Palermo	diesel	5	10	Esso 70	5.8	0.071	0.206	0.189	<0.001	0.15 (−0.15, 0.41)
Palermo	gasoline	1	41	Eni 44	1.8	0.441	0.305	0.299	0.005	0.01 (−0.06, 0.09)
Palermo	gasoline	2	27	Eni 93	4.2	0.034	0.489	0.033	<0.001	0.02 (−0.11, 0.12)
Palermo	gasoline	3	9	Esso 44	7.8	0.004	0.054	0.059	0.006	0.08 (−0.32, 0.42)
Palermo	gasoline	4	44	Esso 41	3.3	0.024	0.286	0.012	<0.001	−0.02 (−0.11, 0.09)
Palermo	gasoline	5	26	Q8 88	4.1	0.096	0.023	0.175	<0.001	0.22 (0.11, 0.37)
Catania	diesel	1	8	Q8 38	3.9	<0.001	0.010	0.035	<0.001	0.35 (0.11, 0.67)
Catania	diesel	2	40	Esso 42	4.1	<0.001	<0.001	0.010	<0.001	0.24 (0.14, 0.32)
Catania	diesel	3	28	Q8 46	3.1	0.002	<0.001	0.843	<0.001	0.18 (0.05, 0.30)
Catania	diesel	4	26	Eni 100	2.9	0.018	0.007	0.044	0.194	0.08 (−0.03, 0.21)
Catania	diesel	5	6	Eni 67	3.1	0.008	0.405	0.095	<0.001	0.37 (−0.12, 0.79)
Catania	gasoline	1	29	Esso 45	3.1	0.792	0.667	0.964	0.788	0.19 (−0.02, 0.50)
Catania	gasoline	2	8	Eni 50	3.4	0.219	0.450	0.420	<0.001	0.09 (−0.10, 0.30)
Catania	gasoline	3	37	Eni 73	4.2	<0.001	0.241	0.185	0.003	0.05 (−0.03, 0.17)
Catania	gasoline	4	34	Q8 35	3.1	<0.001	0.252	0.096	0.503	0.10 (−0.01, 0.21)
Messina	diesel	1	32	Esso 50	4.7	<0.001	0.003	0.642	<0.001	0.08 (0.01, 0.15)
Messina	diesel	2	20	Eni 95	4.3	0.006	0.046	0.379	<0.001	−0.11 (−0.25, 0.04)
Messina	diesel	3	5	Q8 60	6.7	<0.001	<0.001	0.002	<0.001	0.54 (−0.13, 1.02)
Messina	diesel	4	20	Q8 55	6.2	0.010	0.119	<0.001	0.019	0.02 (−0.05, 0.09)
Messina	diesel	5	14	Eni 79	8.6	0.085	0.448	0.134	0.001	0.04 (−0.10, 0.15)
Messina	gasoline	1	32	Eni 69	6.9	0.116	0.055	0.684	0.006	0.16 (0.01, 0.28)
Messina	gasoline	2	5	Q8 80	5.0	<0.001	<0.001	0.039	<0.001	0.08 (−0.18, 0.38)
Messina	gasoline	3	7	Q8 57	6.8	0.001	0.208	0.029	<0.001	0.24 (−0.05, 0.73)
Messina	gasoline	4	22	Esso 64	4.8	<0.001	0.002	0.167	0.065	0.12 (0.04, 0.21)
Messina	gasoline	5	25	Eni 32	4.8	0.117	0.024	0.763	<0.001	0.15 (0.06, 0.25)

Setting 2: Y an independent station, X the nearest station of a major brand

Table C3: The regimes of setting 2. Brand: the most frequent brand of the stations Y of the regime, with its share; X: median distance to the neighbor, in kilometers; p-values of the bootstrap Wald test of the eight lags of X, of its four lags on the same fuel, of its four lags on the other fuel, and of the four lags of the other fuel of Y; long-run pass-through from X on the same fuel, with its bootstrap 95% interval.

city	fuel	regime	pumps	brand (%)	X (km)	bootstrap Wald, p-value				long-run pass-through
						X	X same	X other	Y other	from X, same fuel
Turin	diesel	1	10	indep. 100	1.1	0.001	0.334	0.006	<0.001	0.02 (−0.04, 0.08)
Turin	diesel	2	5	indep. 100	1.1	<0.001	<0.001	0.515	<0.001	0.55 (0.32, 0.71)
Turin	diesel	3	8	indep. 100	1.6	<0.001	<0.001	<0.001	<0.001	0.03 (−0.13, 0.18)
Turin	gasoline	1	7	indep. 100	1.8	<0.001	<0.001	0.913	<0.001	0.27 (0.13, 0.42)
Turin	gasoline	2	5	indep. 100	1.1	0.001	0.002	<0.001	<0.001	−0.05 (−0.15, 0.05)
Turin	gasoline	3	6	indep. 100	1.2	0.087	0.081	0.828	0.321	0.13 (−0.01, 0.23)
Turin	gasoline	4	5	indep. 100	1.1	0.002	0.005	0.551	<0.001	0.12 (−0.23, 0.38)
Milan	diesel	1	6	indep. 100	0.8	0.001	0.248	0.002	0.016	−0.09 (−0.34, 0.11)
Milan	diesel	2	5	indep. 100	0.4	0.024	0.249	0.176	<0.001	0.18 (−0.04, 0.37)
Milan	diesel	3	6	indep. 100	0.9	0.010	0.002	0.147	0.003	0.05 (−0.05, 0.14)
Milan	gasoline	1	12	indep. 100	0.5	0.069	0.087	0.094	<0.001	0.16 (0.05, 0.28)
Milan	gasoline	2	5	indep. 100	0.6	<0.001	<0.001	<0.001	0.241	0.39 (0.15, 0.56)
Venice	diesel	1	8	indep. 100	2.2	<0.001	<0.001	0.894	0.116	0.07 (−0.02, 0.18)
Venice	gasoline	1	8	indep. 100	2.2	<0.001	<0.001	0.050	<0.001	0.20 (0.08, 0.37)
Bologna	diesel	1	13	indep. 100	1.6	0.060	0.200	0.068	<0.001	0.01 (−0.09, 0.09)
Bologna	diesel	2	5	indep. 100	0.6	0.053	0.817	0.207	<0.001	−0.01 (−0.08, 0.06)
Bologna	diesel	3	5	indep. 100	3.5	<0.001	<0.001	0.624	<0.001	−0.02 (−0.20, 0.14)
Bologna	gasoline	1	5	indep. 100	0.6	0.160	0.457	0.535	0.246	0.06 (−0.03, 0.12)
Bologna	gasoline	2	13	indep. 100	2.0	0.004	0.035	0.016	<0.001	0.11 (0.02, 0.19)
Bologna	gasoline	3	5	indep. 100	0.8	<0.001	0.343	<0.001	0.030	0.16 (−0.18, 0.41)
Florence	diesel	1	6	indep. 100	0.8	<0.001	0.250	0.012	0.048	0.14 (−0.08, 0.31)
Florence	gasoline	1	6	indep. 100	0.8	<0.001	<0.001	0.084	0.797	0.23 (0.08, 0.42)
Rome	diesel	1	20	indep. 100	0.8	0.041	0.108	0.114	0.245	0.07 (−0.01, 0.15)
Rome	diesel	2	9	indep. 100	0.6	0.960	0.880	0.988	0.313	0.01 (−0.17, 0.21)
Rome	diesel	3	5	indep. 100	1.2	0.482	0.288	0.435	0.072	−0.21 (−1.08, 0.32)
Rome	diesel	4	5	indep. 100	0.6	0.006	0.102	0.819	<0.001	−0.47 (−1.40, 0.08)
Rome	diesel	5	11	indep. 100	0.5	0.358	0.189	0.635	<0.001	0.06 (−0.20, 0.27)
Rome	gasoline	1	16	indep. 100	0.6	0.951	0.762	0.719	0.008	0.03 (−0.27, 0.21)
Rome	gasoline	2	15	indep. 100	0.7	0.673	0.415	0.953	0.080	0.11 (−0.10, 0.17)
Rome	gasoline	3	8	indep. 100	0.7	0.085	0.038	0.138	0.027	−0.14 (−0.49, 0.12)
Rome	gasoline	4	5	indep. 100	1.1	0.123	0.073	0.139	<0.001	0.48 (−0.29, 1.28)
Rome	gasoline	5	6	indep. 100	0.3	<0.001	0.239	0.004	<0.001	−0.09 (−1.20, 0.41)
Naples	diesel	1	37	indep. 100	0.9	0.338	0.685	0.620	<0.001	0.00 (−0.07, 0.11)
Naples	diesel	2	12	indep. 100	0.9	<0.001	<0.001	<0.001	<0.001	0.19 (0.04, 0.34)
Naples	diesel	3	17	indep. 100	0.5	0.003	0.006	0.068	0.449	0.27 (0.05, 0.39)
Naples	diesel	4	5	indep. 100	0.7	0.211	0.342	0.275	0.005	0.12 (−0.23, 0.58)
Naples	gasoline	1	28	indep. 100	0.5	0.363	0.568	0.316	<0.001	−0.09 (−0.25, 0.06)
Naples	gasoline	2	5	indep. 100	0.8	0.131	0.230	0.870	<0.001	−0.05 (−1.26, 0.65)
Naples	gasoline	3	10	indep. 100	0.8	0.965	0.719	0.908	0.855	0.04 (−0.47, 0.18)
Naples	gasoline	4	28	indep. 100	0.8	0.015	0.131	0.343	<0.001	0.04 (−0.04, 0.10)
Bari	diesel	1	5	indep. 100	0.9	0.030	0.916	0.024	<0.001	0.00 (−0.33, 0.20)
Bari	diesel	2	8	indep. 100	1.1	<0.001	<0.001	0.078	0.004	0.11 (0.02, 0.20)
Bari	gasoline	1	5	indep. 100	0.9	0.131	0.032	0.202	<0.001	0.22 (0.06, 0.39)
Bari	gasoline	2	8	indep. 100	0.9	<0.001	0.019	0.209	0.359	0.20 (0.07, 0.34)
Palermo	diesel	1	14	indep. 100	0.6	<0.001	0.111	<0.001	<0.001	0.13 (−0.08, 0.34)
Palermo	gasoline	1	9	indep. 100	0.5	<0.001	<0.001	0.128	0.079	0.26 (0.10, 0.39)
Palermo	gasoline	2	5	indep. 100	0.6	0.356	0.550	0.116	<0.001	0.11 (−0.06, 0.34)
Catania	diesel	1	6	indep. 100	1.1	0.053	0.019	0.344	0.030	0.17 (−0.04, 0.33)
Catania	diesel	2	6	indep. 100	1.3	0.603	0.383	0.289	0.293	−0.04 (−0.22, 0.11)
Catania	diesel	3	5	indep. 100	2.5	0.129	0.105	0.821	0.081	−0.27 (−0.58, 0.08)
Catania	diesel	4	8	indep. 100	1.1	0.994	0.974	0.934	0.035	0.06 (−0.19, 0.19)
Catania	gasoline	1	5	indep. 100	1.3	0.386	0.249	0.865	<0.001	0.02 (−0.60, 0.65)
Catania	gasoline	2	9	indep. 100	1.1	0.038	0.339	0.014	0.227	0.16 (0.00, 0.28)
Catania	gasoline	3	6	indep. 100	1.3	0.426	0.177	0.345	<0.001	0.16 (−0.03, 0.31)
Catania	gasoline	4	5	indep. 100	1.4	0.587	0.769	0.226	0.023	0.01 (−0.16, 0.32)
Messina	diesel	1	9	indep. 100	1.6	0.032	0.217	0.030	<0.001	0.07 (−0.25, 0.38)
Messina	diesel	2	5	indep. 100	0.9	0.002	0.589	0.180	0.003	−0.07 (−0.48, 0.27)
Messina	gasoline	1	6	indep. 100	2.5	0.001	0.008	<0.001	0.070	0.39 (−0.11, 0.91)

city	fuel	regime	pumps	brand (%)	X (km)	bootstrap Wald, p-value				long-run pass-through
						X	X same	X other	Y other	from X, same fuel
Messina	gasoline	2	8	indep. 100	1.2	<0.001	0.008	0.277	0.002	0.42 (0.21, 0.64)

Setting 3: Y a station of a major brand, X the nearest station of another major brand

Table C4: The regimes of setting 3. Brand: the most frequent brand of the stations Y of the regime, with its share; X: median distance to the neighbor, in kilometers; p-values of the bootstrap Wald test of the eight lags of X, of its four lags on the same fuel, of its four lags on the other fuel, and of the four lags of the other fuel of Y; long-run pass-through from X on the same fuel, with its bootstrap 95% interval.

city	fuel	regime	pumps	brand (%)	X (km)	bootstrap Wald, p-value				long-run pass-through
						X	X same	X other	Y other	from X, same fuel
Turin	diesel	1	37	Tamoil 97	1.4	<0.001	<0.001	0.003	<0.001	0.08 (0.04, 0.13)
Turin	diesel	2	56	Eni 100	0.7	0.048	0.098	0.515	<0.001	−0.01 (−0.08, 0.09)
Turin	diesel	3	135	Api-Ip 37	0.8	<0.001	<0.001	<0.001	<0.001	0.17 (0.14, 0.20)
Turin	diesel	4	45	Api-Ip 42	1.2	<0.001	<0.001	0.518	<0.001	0.20 (0.10, 0.32)
Turin	diesel	5	59	Q8 59	0.9	<0.001	<0.001	0.015	<0.001	0.10 (0.02, 0.14)
Turin	gasoline	1	49	Q8 35	0.9	0.079	0.327	0.031	<0.001	0.06 (0.00, 0.13)
Turin	gasoline	2	52	Esso 37	1.2	<0.001	<0.001	0.003	<0.001	0.20 (0.02, 0.32)
Turin	gasoline	3	123	Api-Ip 39	0.7	<0.001	<0.001	0.016	<0.001	0.16 (0.11, 0.20)
Turin	gasoline	4	41	Eni 100	0.8	<0.001	0.006	0.287	<0.001	0.16 (0.07, 0.26)
Turin	gasoline	5	67	Tamoil 49	1.0	<0.001	<0.001	0.023	<0.001	0.05 (−0.01, 0.12)
Milan	diesel	1	103	Api-Ip 63	0.7	<0.001	<0.001	0.023	<0.001	0.21 (0.16, 0.25)
Milan	diesel	2	33	Api-Ip 27	0.5	<0.001	<0.001	0.014	<0.001	0.26 (0.08, 0.57)
Milan	diesel	3	74	Eni 100	0.7	<0.001	0.053	0.176	<0.001	0.05 (−0.05, 0.15)
Milan	diesel	4	97	Q8 47	0.5	<0.001	<0.001	<0.001	<0.001	0.10 (0.07, 0.13)
Milan	diesel	5	61	Q8 85	0.8	<0.001	<0.001	0.526	<0.001	0.13 (0.08, 0.18)
Milan	gasoline	1	74	Api-Ip 84	0.7	<0.001	<0.001	0.097	<0.001	0.24 (0.16, 0.29)
Milan	gasoline	2	35	Q8 31	0.7	0.008	0.031	0.052	<0.001	0.20 (0.01, 0.38)
Milan	gasoline	3	103	Q8 42	0.5	<0.001	<0.001	0.006	<0.001	0.17 (0.12, 0.22)
Milan	gasoline	4	102	Q8 56	0.7	<0.001	<0.001	0.104	<0.001	0.22 (0.17, 0.25)
Milan	gasoline	5	54	Eni 100	0.7	<0.001	<0.001	0.023	<0.001	0.10 (0.00, 0.29)
Venice	diesel	1	32	Eni 72	2.0	<0.001	<0.001	0.029	<0.001	0.16 (0.05, 0.27)
Venice	diesel	2	15	Eni 47	1.3	0.005	0.111	0.003	<0.001	0.19 (0.01, 0.30)
Venice	diesel	3	22	Q8 64	1.4	0.009	0.017	0.003	<0.001	0.00 (−0.11, 0.10)
Venice	diesel	4	23	Api-Ip 52	1.9	<0.001	0.013	0.204	<0.001	0.10 (0.03, 0.21)
Venice	diesel	5	8	Api-Ip 50	2.5	<0.001	0.002	<0.001	<0.001	0.23 (−0.05, 0.52)
Venice	gasoline	1	37	Q8 38	1.4	<0.001	<0.001	0.679	<0.001	0.21 (0.10, 0.32)
Venice	gasoline	2	12	Esso 67	1.3	0.832	0.948	0.815	0.110	0.02 (−0.07, 0.39)
Venice	gasoline	3	10	Eni 50	2.5	0.014	0.092	0.678	0.002	0.28 (0.08, 0.53)
Venice	gasoline	4	21	Eni 100	2.6	0.026	0.846	0.055	<0.001	0.04 (−0.12, 0.19)
Venice	gasoline	5	20	Api-Ip 55	1.8	<0.001	0.020	0.133	0.118	−0.01 (−0.10, 0.11)
Bologna	diesel	1	28	Eni 100	0.9	<0.001	0.130	0.005	0.018	0.15 (−0.02, 0.31)
Bologna	diesel	2	6	Eni 67	0.9	0.491	0.518	0.204	<0.001	0.16 (−0.13, 0.44)
Bologna	diesel	3	51	Api-Ip 57	0.8	<0.001	<0.001	0.018	<0.001	0.12 (0.07, 0.18)
Bologna	diesel	4	32	Q8 59	1.4	<0.001	<0.001	0.003	<0.001	0.17 (0.09, 0.25)
Bologna	diesel	5	39	Esso 33	1.5	<0.001	<0.001	0.208	0.037	0.17 (0.09, 0.23)
Bologna	gasoline	1	41	Q8 44	1.4	<0.001	<0.001	0.100	0.005	0.05 (−0.06, 0.15)
Bologna	gasoline	2	22	Q8 45	1.3	0.023	0.131	0.177	0.011	−0.13 (−0.24, 0.01)
Bologna	gasoline	3	19	Eni 100	0.7	0.001	0.012	0.260	0.109	0.25 (0.09, 0.38)
Bologna	gasoline	4	64	Api-Ip 50	0.8	<0.001	<0.001	0.007	<0.001	0.21 (0.15, 0.26)
Bologna	gasoline	5	10	Eni 40	1.4	<0.001	<0.001	0.254	<0.001	0.30 (0.03, 0.57)
Florence	diesel	1	8	Q8 62	3.3	0.144	0.164	0.322	0.068	−0.20 (−0.61, 0.23)
Florence	diesel	2	38	Q8 53	0.9	<0.001	<0.001	0.004	<0.001	0.19 (0.11, 0.25)
Florence	diesel	3	56	Q8 39	0.6	<0.001	<0.001	0.011	0.397	0.23 (0.17, 0.29)
Florence	diesel	4	8	Eni 50	1.1	<0.001	<0.001	0.844	0.014	0.44 (0.21, 0.65)
Florence	diesel	5	33	Eni 100	0.7	<0.001	0.001	0.249	<0.001	0.18 (0.08, 0.29)
Florence	gasoline	1	12	Q8 33	1.1	<0.001	<0.001	0.814	<0.001	0.26 (0.07, 0.46)

city	fuel	regime	pumps	brand (%)	X (km)	bootstrap Wald, p-value				long-run pass-through
						X	X same	X other	Y other	from X, same fuel
Florence	gasoline	2	52	Api-Ip 38	0.6	<0.001	<0.001	0.562	<0.001	0.15 (0.09, 0.22)
Florence	gasoline	3	16	Q8 50	1.9	<0.001	<0.001	0.453	0.731	0.30 (0.14, 0.43)
Florence	gasoline	4	35	Q8 54	1.0	<0.001	0.001	0.097	<0.001	0.10 (−0.02, 0.17)
Florence	gasoline	5	28	Eni 100	0.7	0.010	0.315	0.381	0.083	0.09 (−0.06, 0.20)
Rome	diesel	1	62	Api-Ip 50	1.1	<0.001	<0.001	0.043	<0.001	0.16 (0.07, 0.23)
Rome	diesel	2	177	Eni 64	0.9	<0.001	<0.001	0.199	<0.001	0.13 (0.08, 0.17)
Rome	diesel	3	85	Q8 100	0.9	0.154	0.161	0.962	0.770	0.10 (0.03, 0.22)
Rome	diesel	4	200	Api-Ip 62	0.7	<0.001	<0.001	0.129	<0.001	0.14 (0.11, 0.16)
Rome	diesel	5	27	Api-Ip 41	0.6	0.759	0.617	0.708	0.256	0.14 (−0.04, 0.32)
Rome	gasoline	1	55	Api-Ip 45	0.9	<0.001	<0.001	0.274	<0.001	0.13 (0.03, 0.24)
Rome	gasoline	2	215	Api-Ip 41	0.8	<0.001	<0.001	<0.001	<0.001	0.16 (0.12, 0.19)
Rome	gasoline	3	135	Q8 41	0.9	<0.001	<0.001	0.003	<0.001	0.13 (0.10, 0.17)
Rome	gasoline	4	5	Q8 60	2.1	0.700	0.438	0.832	0.493	0.22 (−0.34, 0.60)
Rome	gasoline	5	141	Api-Ip 48	0.6	<0.001	<0.001	0.004	<0.001	0.17 (0.12, 0.22)
Naples	diesel	1	32	Eni 34	1.0	0.003	0.118	0.105	<0.001	0.09 (−0.02, 0.22)
Naples	diesel	2	58	Q8 50	1.0	<0.001	<0.001	0.007	<0.001	0.11 (0.06, 0.15)
Naples	diesel	3	51	Q8 47	0.6	<0.001	<0.001	0.122	0.139	0.07 (−0.01, 0.15)
Naples	diesel	4	39	Eni 97	0.8	0.538	0.987	0.698	<0.001	−0.09 (−0.23, 0.14)
Naples	diesel	5	51	Api-Ip 47	0.7	0.436	0.111	0.865	0.494	0.18 (−0.04, 0.26)
Naples	gasoline	1	29	Eni 100	0.7	0.749	0.989	0.958	0.005	0.09 (−0.06, 0.30)
Naples	gasoline	2	24	Api-Ip 33	1.0	<0.001	<0.001	<0.001	<0.001	0.22 (0.07, 0.41)
Naples	gasoline	3	62	Q8 47	1.0	<0.001	<0.001	0.004	<0.001	0.13 (0.08, 0.19)
Naples	gasoline	4	50	Esso 32	0.8	<0.001	0.248	0.422	0.004	0.04 (−0.04, 0.32)
Naples	gasoline	5	66	Api-Ip 36	0.6	<0.001	<0.001	0.132	<0.001	0.14 (0.05, 0.17)
Bari	diesel	1	51	Q8 57	1.1	<0.001	<0.001	0.363	<0.001	0.11 (0.07, 0.16)
Bari	diesel	2	26	Eni 38	1.0	0.011	0.001	0.021	<0.001	0.07 (−0.05, 0.21)
Bari	diesel	3	44	Eni 100	1.0	<0.001	0.354	0.131	<0.001	0.05 (−0.03, 0.12)
Bari	diesel	4	5	Api-Ip 80	0.6	0.173	0.460	0.164	0.007	0.63 (−0.11, 1.51)
Bari	diesel	5	74	Api-Ip 80	1.2	<0.001	<0.001	0.041	0.003	0.08 (0.05, 0.13)
Bari	gasoline	1	75	Q8 53	1.0	<0.001	<0.001	0.181	<0.001	0.21 (0.16, 0.28)
Bari	gasoline	2	5	Api-Ip 60	0.5	0.061	0.352	0.526	<0.001	0.29 (−0.03, 0.62)
Bari	gasoline	3	53	Eni 96	1.0	<0.001	<0.001	0.007	<0.001	0.15 (0.09, 0.20)
Bari	gasoline	4	13	Api-Ip 46	1.0	0.003	0.044	0.039	0.141	0.49 (0.00, 1.23)
Bari	gasoline	5	54	Api-Ip 91	1.2	<0.001	<0.001	0.006	<0.001	0.20 (0.16, 0.26)
Palermo	diesel	1	67	Q8 40	0.6	<0.001	<0.001	<0.001	<0.001	0.17 (0.12, 0.21)
Palermo	diesel	2	45	Eni 31	0.5	<0.001	<0.001	<0.001	<0.001	0.14 (0.08, 0.23)
Palermo	diesel	3	13	Esso 62	2.9	0.006	0.114	0.022	<0.001	0.23 (0.02, 0.37)
Palermo	diesel	4	22	Eni 100	0.6	0.322	0.269	0.781	0.075	0.17 (0.02, 0.30)
Palermo	gasoline	1	29	Q8 86	1.3	0.321	0.170	0.244	0.008	0.12 (0.00, 0.21)
Palermo	gasoline	2	8	Esso 50	4.0	<0.001	0.164	0.044	0.007	0.17 (−0.22, 0.54)
Palermo	gasoline	3	37	Esso 43	0.4	<0.001	<0.001	0.137	<0.001	0.20 (0.11, 0.29)
Palermo	gasoline	4	41	Eni 44	0.5	<0.001	0.011	0.055	0.005	0.12 (0.05, 0.20)
Palermo	gasoline	5	32	Eni 78	0.6	0.263	0.601	0.607	0.003	0.02 (−0.09, 0.12)
Catania	diesel	1	26	Eni 100	0.7	0.017	0.262	0.724	0.843	−0.01 (−0.08, 0.33)
Catania	diesel	2	10	Eni 30	1.5	0.637	0.449	0.337	<0.001	0.10 (−0.04, 0.29)
Catania	diesel	3	5	Eni 60	0.6	0.477	0.174	0.589	0.366	0.34 (−0.07, 1.01)
Catania	diesel	4	28	Q8 46	0.7	<0.001	<0.001	<0.001	<0.001	0.18 (0.04, 0.27)
Catania	diesel	5	39	Esso 44	0.9	<0.001	<0.001	0.009	0.005	0.32 (0.25, 0.38)
Catania	gasoline	1	40	Eni 68	1.0	<0.001	<0.001	0.057	<0.001	0.14 (0.09, 0.19)
Catania	gasoline	2	23	Esso 57	1.1	<0.001	0.002	0.142	0.783	0.19 (0.08, 0.30)
Catania	gasoline	3	5	Eni 60	1.5	0.394	0.148	0.195	0.001	0.35 (0.07, 0.65)
Catania	gasoline	4	8	Q8 38	0.8	0.723	0.526	0.515	0.741	0.20 (−0.13, 0.56)
Catania	gasoline	5	32	Q8 38	0.5	<0.001	<0.001	0.432	0.059	0.10 (0.03, 0.17)
Messina	diesel	1	9	Q8 56	2.0	<0.001	0.219	0.067	<0.001	0.18 (−0.05, 0.43)
Messina	diesel	2	14	Eni 57	0.8	0.002	0.004	0.006	<0.001	0.08 (−0.09, 0.27)
Messina	diesel	3	20	Q8 60	1.2	<0.001	0.053	<0.001	0.060	0.16 (0.05, 0.27)
Messina	diesel	4	23	Eni 91	0.9	<0.001	0.004	0.097	<0.001	0.18 (0.09, 0.30)
Messina	diesel	5	25	Esso 64	1.1	<0.001	<0.001	0.004	<0.001	0.30 (0.17, 0.39)
Messina	gasoline	1	16	Eni 50	0.8	<0.001	0.011	0.363	0.008	0.24 (0.08, 0.42)
Messina	gasoline	2	26	Esso 42	1.2	<0.001	<0.001	<0.001	0.006	0.32 (0.25, 0.40)
Messina	gasoline	3	16	Q8 50	1.4	<0.001	<0.001	0.008	<0.001	0.39 (0.23, 0.53)
Messina	gasoline	4	28	Eni 79	0.9	<0.001	0.019	0.004	0.133	0.22 (0.08, 0.37)
Messina	gasoline	5	5	Q8 80	2.0	<0.001	0.264	0.001	<0.001	0.42 (−0.08, 0.84)

All the regimes at once

Table C5: The Wald test of the eight lags of X in all the regimes of a model at once: statistic and degrees of freedom; p-value against the chi-square and against the bootstrap distribution of the statistic centered at the estimate (its smallest value is 1/101); p-value of the same test on the pooled STEM model.

setting	city	fuel	regimes	W (df)	p_{χ^2}	p_{boot}	pooled p
1	Turin	diesel	5	202.6 (40)	<0.001	0.010	0.017
1	Turin	gasoline	5	169.3 (40)	<0.001	0.010	0.026
1	Milan	diesel	5	281.4 (40)	<0.001	0.010	<0.001
1	Milan	gasoline	4	187.5 (32)	<0.001	0.010	<0.001
1	Venice	diesel	5	244.1 (40)	<0.001	0.010	0.081
1	Venice	gasoline	5	257.2 (40)	<0.001	0.010	0.587
1	Bologna	diesel	4	261.6 (32)	<0.001	0.010	0.030
1	Bologna	gasoline	5	154.8 (40)	<0.001	0.010	0.588
1	Florence	diesel	5	141.6 (40)	<0.001	0.010	0.507
1	Florence	gasoline	5	187.5 (40)	<0.001	0.010	0.060
1	Rome	diesel	4	105.8 (32)	<0.001	0.010	<0.001
1	Rome	gasoline	5	231.1 (40)	<0.001	0.010	<0.001
1	Naples	diesel	5	252.4 (40)	<0.001	0.010	<0.001
1	Naples	gasoline	5	259.6 (40)	<0.001	0.010	<0.001
1	Bari	diesel	4	59.3 (32)	0.002	0.050	0.088
1	Bari	gasoline	5	102.0 (40)	<0.001	0.010	0.338
1	Palermo	diesel	5	112.4 (40)	<0.001	0.010	<0.001
1	Palermo	gasoline	5	105.5 (40)	<0.001	0.010	0.002
1	Catania	diesel	5	199.3 (40)	<0.001	0.010	<0.001
1	Catania	gasoline	4	117.0 (32)	<0.001	0.010	<0.001
1	Messina	diesel	5	213.2 (40)	<0.001	0.010	<0.001
1	Messina	gasoline	5	194.6 (40)	<0.001	0.010	0.035
2	Turin	diesel	3	111.0 (24)	<0.001	0.010	0.070
2	Turin	gasoline	4	145.8 (32)	<0.001	0.010	<0.001
2	Milan	diesel	3	85.1 (24)	<0.001	0.010	0.158
2	Milan	gasoline	2	73.7 (16)	<0.001	0.010	<0.001
2	Venice	diesel	1	30.2 (8)	<0.001	0.010	<0.001
2	Venice	gasoline	1	62.9 (8)	<0.001	0.010	<0.001
2	Bologna	diesel	3	90.7 (24)	<0.001	0.010	<0.001
2	Bologna	gasoline	3	102.5 (24)	<0.001	0.010	<0.001
2	Florence	diesel	1	51.0 (8)	<0.001	0.010	<0.001
2	Florence	gasoline	1	47.2 (8)	<0.001	0.010	<0.001
2	Rome	diesel	5	69.8 (40)	0.002	0.030	0.236
2	Rome	gasoline	5	133.8 (40)	<0.001	0.010	0.105
2	Naples	diesel	4	162.5 (32)	<0.001	0.010	<0.001
2	Naples	gasoline	4	55.7 (32)	0.006	0.020	0.051
2	Bari	diesel	2	53.2 (16)	<0.001	0.010	0.158
2	Bari	gasoline	2	50.8 (16)	<0.001	0.010	<0.001
2	Palermo	diesel	1	35.3 (8)	<0.001	0.010	<0.001
2	Palermo	gasoline	2	66.1 (16)	<0.001	0.010	<0.001
2	Catania	diesel	4	46.1 (32)	0.050	0.129	0.737
2	Catania	gasoline	4	56.2 (32)	0.005	0.030	0.055
2	Messina	diesel	2	47.4 (16)	<0.001	0.010	0.016
2	Messina	gasoline	2	59.4 (16)	<0.001	0.010	<0.001

setting	city	fuel	regimes	W (df)	p_{χ^2}	p_{boot}	pooled p
3	Turin	diesel	5	857.5 (40)	<0.001	0.010	<0.001
3	Turin	gasoline	5	585.7 (40)	<0.001	0.010	<0.001
3	Milan	diesel	5	878.1 (40)	<0.001	0.010	<0.001
3	Milan	gasoline	5	756.0 (40)	<0.001	0.010	<0.001
3	Venice	diesel	5	236.2 (40)	<0.001	0.010	<0.001
3	Venice	gasoline	5	176.3 (40)	<0.001	0.010	<0.001
3	Bologna	diesel	5	435.9 (40)	<0.001	0.010	<0.001
3	Bologna	gasoline	5	607.7 (40)	<0.001	0.010	<0.001
3	Florence	diesel	5	587.1 (40)	<0.001	0.010	<0.001
3	Florence	gasoline	5	322.4 (40)	<0.001	0.010	<0.001
3	Rome	diesel	5	988.4 (40)	<0.001	0.010	<0.001
3	Rome	gasoline	5	1180.4 (40)	<0.001	0.010	<0.001
3	Naples	diesel	5	317.3 (40)	<0.001	0.010	<0.001
3	Naples	gasoline	5	523.0 (40)	<0.001	0.010	<0.001
3	Bari	diesel	5	276.1 (40)	<0.001	0.010	<0.001
3	Bari	gasoline	5	514.3 (40)	<0.001	0.010	<0.001
3	Palermo	diesel	4	249.4 (32)	<0.001	0.010	<0.001
3	Palermo	gasoline	5	297.3 (40)	<0.001	0.010	<0.001
3	Catania	diesel	5	186.9 (40)	<0.001	0.010	<0.001
3	Catania	gasoline	5	260.5 (40)	<0.001	0.010	<0.001
3	Messina	diesel	5	364.3 (40)	<0.001	0.010	<0.001
3	Messina	gasoline	5	612.0 (40)	<0.001	0.010	<0.001

C5 The pump-by-pump benchmark

For every station Y the regression (C2) without the latent process, the spatial error and the regimes is estimated by least squares on its own weeks, and the eight lags of X are tested in two ways: by the classical F test, with the covariance $s^2(Z'Z)^{-1}$, valid when the error variance is constant, and by the Wald test with the heteroskedasticity-consistent covariance HC1 (White, 1980; MacKinnon and White, 1985). Over the cities, the F test rejects for 40 to 85% of the stations and the HC1 test for 17 to 67%. With sticky prices the variance of the changes is far from constant, most weeks without a change and a few large ones around the cost shocks, and the F test reads those few weeks as full information; the HC1 test is the one to read. The combination over the stations of a city by the test of Dumitrescu and Hurlin (2012) rejects non-causality in all the 66 models, for both versions.

Table C6 shows that the HC1 rejections grow as X gets closer: in setting 3 from 31% between two and ten kilometers to 42% within 500 meters, in setting 1 from 29–31% to 40%.

Table C6: Share of the stations Y for which the least-squares Granger test of the eight lags of X rejects at 5%, by the distance to X and setting: F test and HC1 test.

	setting 1			setting 2			setting 3		
distance to X (km)	pumps	F	HC1	pumps	F	HC1	pumps	F	HC1
0–0.5	202	57	40	142	63	31	1516	62	42
0.5–1	364	57	35	136	57	33	1232	59	38
1–2	790	56	33	122	59	43	1154	56	34
2–5	1726	50	30	104	43	29	724	55	31
5–10	1062	50	31	16	38	19	182	52	31
>10	708	47	29	6	17	33	46	67	41

C6 Diagnostics and robustness

Three features of the data call for care in reading single estimates. The cap of five regimes binds in most models of settings 1 and 3 (Section C3), so the BIC would split the stations further. Sticky prices give errors with heavy tails and strong conditional heteroskedasticity, so the Gaussian model and its parametric bootstrap are approximations, whose effect on the size of the tests is not known. The likelihood of the largest regimes has more than one local maximum, but the tests of leadership proved robust to it. This section documents the last two points.

Residuals. On the residuals of the selected fits from their fitted signal, station by station: the Ljung–Box test at lag 4 rejects for 40 to 44% of the stations (medians over the regimes), the same test on the squared residuals for 78 to 81%, and the median kurtosis is 10 to 11, while the first-order autocorrelation is near zero. The model leaves little linear dependence but strong conditional heteroskedasticity and heavy tails, the signature of sticky prices: weeks without a change alternate with large adjustments. The Gaussian model and its parametric bootstrap are therefore approximations; the effect of the non-Gaussian errors on the size of the bootstrap tests is not known.

The range of the spatial field. The range parameter of 56 of the 273 regimes is at the upper limit of the Newton–Raphson step, where the spatial correlation has fallen below 0.05 already between the two closest stations of the regime, and that of 3 at the lower limit, where it stays above 0.95 between the two farthest; 35 of these 59 are regimes of at most ten stations, where a range cannot be identified (Section A1.10 of Computational and Algorithmic Supplement A).

Local maxima. The likelihood of a regime can have more than one local maximum, and the EM algorithm reaches the one that its start leads to. Every regime of the 66 selected fits was refitted from least squares, with the EM algorithm stopping only when both the parameters and the log-likelihood had settled, and compared with the selected fit, whose regimes start from the alternation. Of the 268 regimes with more than one start (5 models are pooled), 191 reach the same log-likelihood within one unit, 52 differ by more than 10 and 17 by more than 100; least squares reaches the higher maximum by more than 10 in 38 regimes, the selected fit in 14. The share grows with the size of the regime: 11% of the regimes of at most ten stations, 47% of those of more than one hundred. The two maxima differ in the range of the spatial field and in the persistence of the latent process, not in the regression: in the 52 regimes the lag coefficients move by a median of 0.13 to 0.17 bootstrap standard errors, of the 416 coefficients of the lags of X one moves by more than two, and no coefficient significant at both maxima changes sign. The 5% decision of the Wald test

of leadership changes in 4 regimes: in two the selected fit is at the higher maximum, so its decision stands; in the other two the selected fit is at the lower one, Turin diesel in setting 3 ($p = 0.048$ against 0.050) and a regime of five independents of Catania, gasoline ($p = 0.39$ against 0.027). The estimates and the tests of leadership are therefore kept.

C7 The pass-through of the excise duties

The window contains five changes of the excise duties: the temporary cut of 22 March 2022, its reduction on 1 December 2022 and its end on 1 January 2023, the realignment of 15 May 2025, −1.5 cents per liter on gasoline and +1.5 on diesel, and the equal duties on the two fuels from 1 January 2026. The part of a change that every station passes through at once is removed by the centering (C1), so the pulse of the week of a change measures how far the relative price of a regime is from the national price in that week: a positive coefficient, a regime above the national price, a negative one below it. Table C7 summarizes the coefficients over the regimes. The two large events, the cut and its end, move most regimes, by up to 13 to 16 cents, and the regimes of a city differ by about 25 cents in the week of the cut; in the following week the deviation is smaller, and after the cut it is gone for most regimes. The small events move few regimes, and the realignment of 2025 almost none.

Table C7: The pulses of the changes of the excise duties, over the 273 regimes of the 66 models: share of the regimes whose bootstrap 95% interval excludes zero; 10th percentile, median and 90th percentile of the coefficients, in cents per liter; median over the models of the difference between the largest and the smallest coefficient of the regimes of a city.

fuel	pulse	$\neq 0$ (%)	10%	median	90%	spread in a city
gasoline	excise cut	71	−6.1	−2.3	13.3	25.1
gasoline	excise cut next	32	−3.2	−0.6	1.9	9.6
gasoline	cut reduced	14	−2.4	0.2	1.3	4.6
gasoline	cut reduced next	12	−1.6	0.4	1.3	3.8
gasoline	cut ends	79	−6.0	−3.1	12.3	18.8
gasoline	cut ends next	66	−5.7	1.2	4.0	9.9
gasoline	realignment	1	−0.7	0.1	0.7	1.8
gasoline	realignment next	0	−0.5	0.0	0.5	1.2
gasoline	equal excise	20	−2.1	−0.4	2.2	4.4
gasoline	equal excise next	1	−0.8	−0.1	0.6	3.4
diesel	excise cut	81	−6.4	−3.8	16.1	26.6
diesel	excise cut next	19	−3.0	−1.0	1.0	7.2
diesel	cut reduced	13	−3.0	0.0	1.6	6.4
diesel	cut reduced next	8	−1.5	0.3	1.8	5.1
diesel	cut ends	67	−6.0	−1.4	11.5	18.1
diesel	cut ends next	48	−4.9	1.1	3.2	9.8
diesel	realignment	1	−1.2	0.2	1.0	2.6
diesel	realignment next	0	−0.8	0.0	0.9	1.9
diesel	equal excise	7	−1.6	0.4	1.5	3.8
diesel	equal excise next	2	−1.0	0.2	0.9	2.1

C8 The small samples of the independent stations

Setting 2 has 6 to 71 stations per city, below the smallest network of the simulation study (40 locations) in all cities but Rome and Naples. The procedure behaves as follows.

- Florence and Venice, with 6 and 8 stations, are fitted by the pooled STEM model; in Palermo, diesel, the rule selects the pooled model among $K \leq 2$. Elsewhere $\hat{K}$ is two to five, at the cap $\lfloor n/5 \rfloor$ in 13 of the 22 models, with regimes of five to 37 stations.
- The penalty is zero in 16 models, and 0.5 and 1 in Milan diesel (17 stations) and Bari gasoline (13), where the strong penalties, outside the band of the first step, win the second.
- All the bootstraps have 100 usable draws, and the partition is reproduced in the median draw of every model but Naples, gasoline (median adjusted Rand index 0.66).
- The range of the spatial field is at a bound in 21 of the 59 regimes: a regime of five to ten stations, kilometers apart, cannot identify it.

On real data the behavior can be described, not the properties of the procedure; below 40 stations the simulation study offers no support, and the results of setting 2 are to be read with that in mind.

C9 Replication material

The data pre-treatment, the three main scripts of the application (one per setting), the bootstrap tests and the tables and figures of this supplement and of Section 5 of the manuscript are produced by the replication material of the paper, described in the code supplement of the package.

References

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